

Part 2 of 2

Wireless Sensor Networks

Design Challenges Embedded Platform Selection

While building a wireless sensor network, one faces several challenges that need to be addressed at the design stage itself. Moreover, as new wireless standards emerge, plenty of standards-based embedded products and solutions are available to quickly build and deploy a wireless sensor network. Therefore selecting the right embedded product and its platform is a complex task. This section discusses the software applications that need to be considered

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A wireless sensor network (WSN) is a large-scale distributed architecture that incorporates hardware and software elements. In the first part of this article, carried in August issue of *Electronics For You*, the author had discussed the primary and secondary design challenges of WSN, the design considerations and the embedded platform selection. The author had also talked about the basic features of WSN components, and the factors to consider while choosing them and transmit power—as these decide the range of a radio. This section discusses the software applications that need to be considered.

Software application considerations. A component-based framework is good for developing application software for a WSN. The components of the framework provide the functionality of sensors, sensor nodes and the whole sensor network. Based on these components, applications are classified under sensor applications, node applications and network applications.

As shown in Fig. 3, the sensor application does the essential and basic functions of the local sensor node.

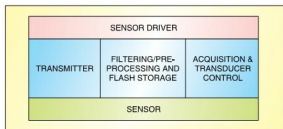


Fig. 3: Structure of a sensor application

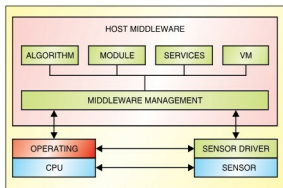


Fig. 4: Structure of a node application

The sensor driver initialises the sensor hardware and performs the measurements in the sensors.

The node application does the application-specific tasks and functions of the middleware to build up and maintain the network, e.g., routing, looking for nodes, and discovering services and

self-localisation.

Fig. 4 shows the structure of a node application. The operating system handles boot-up, initialisation of the hardware, device-specific tasks, scheduling, memory management and process management. The middleware is the superior software layer, which organises the cooperation of the distributed nodes in the network.

The sensor network application describes the main tasks and the required services of the entire network. Fig. 5 shows the logical view of a sensor network application. The nodes can be contacted only through the services of the middleware layers.

The distributed middleware oversees the cooperation of the services within the network. It is logically located in the network layer, but it exists physically in the nodes. All layers together, in conjunction with their configuration, comprise the sensor network application.